

FreeLM: Fine-Tuning-Free Language Model

Hypothesis: A novel fine-tuning-free strategy for language models, which considers both language signal and a unified 'teacher signal', can produce a reasonably sized model capable of generalizing to a wide range of tasks without requiring task-specific fine-tuning. This is achieved by iteratively training the model on both raw language and a unified dataset of tasks formatted as 'proposition correctness judgments'.

Related Work: Pre-trained language models (PLMs) like the GPT series follow a pre-training then fine-tuning paradigm, which is costly for deployment and training. Zero-shot and few-shot approaches have unsatisfactory performance, particularly on understanding tasks, because PLMs are trained only with a language signal that is not task-aware. Instruction-tuning models like InstructGPT and FLAN improve zero-shot performance by making the language signal more task-aware, but adaptation is still necessary for best performance. The research gap is the need for a method to create a reasonably sized model that can achieve good performance on a range of pre-defined tasks with low training and deployment costs, eliminating the need for fine-tuning.

Abstract: [Content omitted for brevity]

Experiments:

Description: To evaluate the language understanding ability of FreeLM in a fine-tuning-free scenario against strong baselines.

Method: FreeLM (a 0.3B parameter model based on GPT-2) will be evaluated on a subset of the GLUE benchmark (CoLA, SST-2, MRPC, QQP, MNLI, QNLI, RTE). The evaluation will be performed in a zero-shot setting. Baselines for comparison will include GPT-2 (zero-shot, few-shot, and fine-tuned), 175B GPT-3 (zero-shot and few-shot), and InstructGPT (zero-shot and few-shot).

Metrics: Matthew's Correlation (CoLA), Accuracy (SST-2, MNLI, QNLI, RTE), F1/Accuracy (MRPC, QQP), Average score across tasks

Description: To evaluate the language generation performance of FreeLM to assess if it retains this capability after the proposed training.

Method: FreeLM will be evaluated against the baseline GPT-2 model. The evaluation will be conducted on three public datasets: LAMBADA, WikiText2, and PTB.

Metrics: Perplexity (PPL)

Description: To analyze the impact of the iterative training strategy by removing the language signal objective.

Method: An ablation study will be conducted by training a version of FreeLM without the language signal objective ('w.o. L.S.'). This model's performance will be compared to the standard FreeLM on the GLUE benchmark tasks. The training loss curves will also be compared.

Metrics: GLUE benchmark metrics (Mat. Corr, Acc, F1), Proposition loss during training

Description: To evaluate the generalization ability of FreeLM on unseen tasks not included in the unified training data.

Method: A new model, FreeLM_U, will be trained by removing four datasets (MRPC, RTE, COPA, and CB) from the unified training data. This model's performance on these four unseen tasks will be compared to the standard FreeLM and a fine-tuned GPT-2 baseline.

Metrics: F1/Accuracy (MRPC), Accuracy (RTE, COPA, CB), Average score across unseen tasks

Description: To qualitatively evaluate the generation ability, specifically text consistency, via a case study on a story cloze task.

Method: FreeLM and GPT-2 will generate text continuations for input prompts from the StoryCloze dataset. To ensure a fair comparison, a new FreeLM model will be trained with the StoryCloze dataset removed from its unified data. The generated text will be manually inspected.

Metrics: Manual inspection for readability, grammatical correctness, and context consistency/logical association.

Risk Factor:

There are studies suggesting that PPL does not fully reflect the generation ability of language models Wang et al. (2022). Nevertheless, there are no good alternative metrics.

We believe that when the size of FreeLM exceeds 1B, task prefix will have little effect.

Meta ICLR criticism: The paper over claims and **doesn't design experiment properly**. FreeLM is claimed as "finetuning free" and compared with GPT-3 and InstructGPT at zero-shot and few-shot setting. However, **training split data of eval tasks (or of similar eval tasks) are used in pretraining**. Thus, FreeLM is trained on much more information (and similar domains to eval tasks) than baselines, which **make the comparison not fair**. This also leads to questions how generalizable to unseen domains/tasks FreeLM will be. The **method is not novel since** the approach is simply a mixture of pretaining and multi-task finetuning. **More experiments and analysis are required** to show how FreeLM surpasses previous techniques or introduces new capability to language models. FreeLM is limited to classification tasks. It's not clear how this approuce can be extended to generative tasks or others. The introduction of SFT signals seems throwing away the main benefit of language models.

Mean ICLR
Soundness Score
1.25/4.0

